

Jugal Gajjar

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RESEARCH SUMMARY

My research focuses on trustworthy and reasoning-centric AI systems, particularly at the intersection of large language models (LLMs), retrieval-augmented generation (RAG), agentic workflows, and human-AI collaboration. I am interested in developing structured and verifiable AI systems that combine neural representations with symbolic reasoning, graph-based knowledge, and grounded validation for reliable decision-making and software intelligence. My work spans structured reasoning and post-training alignment for faithful attribution (ACL 2026), adaptive multi-space knowledge graph embeddings (IEEE BigData 2025), and attribution stability evaluation for robust XAI (CVPR 2026). A parallel thread covers AI for Software Engineering and Code Security, including execution-grounded agentic repair (M.S. Thesis; IEEE ICMLA 2025), hybrid GNN-LLM vulnerability detection (Complex Networks 2025), and language-agnostic malicious code detection (IEEE IEEE 2025). Across both threads, a unifying principle is that reliable AI systems must ground their outputs in verifiable signals rather than relying on generation alone.

EDUCATION

M.S. in Computer Science (Machine Intelligence & Cognition)

The George Washington University, *Washington, DC*

May 2026

GPA: 3.88/4.0

B.Tech. in Computer Science and Engineering

Navrachana University, *Vadodara, India*

May 2024

CGPA: 9.39/10

RESEARCH PUBLICATIONS

Verify Before You Fix: Agentic Execution Grounding for Trustworthy Cross-Language Code Analysis

Solo Author; Under Review at NeurIPS 2026

RSAT: Structured Attribution Makes Small Language Models Faithful Table Reasoners

First Author; SURGeLLM Workshop (Proceedings Track) at ACL 2026

Jul. 2026

Feature Attribution Stability Suite: How Stable Are Post-Hoc Attributions?

Co-Author; XAI4CV Workshop (Proceedings Track) at CVPR 2026

Jun. 2026

Bridging Semantics & Structure for Software Vulnerability Detection using Hybrid Network Models

First Author; 14th International Conference on Complex Networks and their Applications

Dec. 2025

HyperComplex: Adaptive Multi-Space Knowledge Graph Embeddings

First Author; 2025 IEEE International Conference on Big Data (IEEE BigData 2025)

Dec. 2025

SecureFixAgent: A Hybrid LLM Agent for Automated Python Static Vulnerability Repair

First Author; 2025 International Conference on Machine Learning and Applications (ICMLA)

Dec. 2025

MLCPD: A Multi-Language Code Parser Dataset with Universal AST Representation

First Author; arXiv preprint (cs.SE)

Oct. 2025

MalCodeAI: Autonomous Vulnerability Detection and Remediation via Language Agnostic Code Reasoning

First Author; IEEE 26th International Conference on Information Reuse and Integration (IRI 2025)

Aug. 2025

Multimodal Sentiment Analysis on CMU-MOSEI Dataset using Transformer-based Models

First Author; arXiv preprint (cs.CL)

May 2025

PROFESSIONAL SERVICE

Program Committee Member:

- 20th ACM International Conference on Web Search and Data Mining (WSDM)

Feb. 2027

- 6th Workshop on KG and BigData, IEEE Big Data (KGBigdata 2026)

Dec. 2026

- 38 th IEEE International Conference on Tools with Artificial Intelligence (ICTAI)	Nov. 2026
Reviewer:	
- ICMLA 2026	Oct. 2026
- ACL 2026 (GEM Workshop, SURGeLLM Workshop, CDL Workshop)	Jul. 2026
- IEEE USBEREIT 2026	May. 2026
- ICMLA 2025 (ROSE-LLM Special Session)	Dec. 2025

SELECTED RESEARCH PROJECTS

RSAT: Structured Attribution for Faithful Table Reasoning in Small LLMs Dec. 2025 – Mar. 2026

- Designed a two-phase SFT + GRPO training pipeline enabling small language models (1–8B) to generate step-by-step reasoning with explicit cell-level citations.
- Formulated a multi-objective reward function combining faithfulness, citation validity, and reasoning efficiency for GRPO-based policy optimization.
- Applied GRPO-based policy optimization to improve reasoning faithfulness and attribution quality under structured reward signals.
- Achieved 3.7× improvement in attribution faithfulness over supervised fine-tuning baselines.

Cross-Language Vulnerability Detection, Validation, and Repair (M.S. Thesis) Jun. 2025 – Mar. 2026

- Built a uAST-based agentic framework for vulnerability detection and repair, achieving a 69.74% end-to-end resolution rate on a 120K-sample dataset.
- Designed a GNN + transformer fusion model, reaching 74.4–78.2% zero-shot cross-language F1.
- Developed an end-to-end pipeline with execution-based validation, reducing false positives by up to 62% and successfully repairing 81–87% of cases.

VulnGraph: Graph + LLM Embeddings for Vulnerability Detection Jul. 2025 – Sep. 2025

- Engineered a multimodal fusion model integrating AST/CFG graph embeddings with LLM-based semantic representations.
- Integrated structural program representations (AST/CFG) as symbolic constraints to guide neural-LLM feature fusion via a two-way gating mechanism for effective alignment.
- Achieved 93.57% accuracy, outperforming GNN-only (+8.36%) and LLM-only (+17.81%) baselines with interpretable outputs.

MLCPD: Multi-Language Code Parser Dataset with Universal AST Schema May 2025 – Oct. 2025

- Designed a four-layer universal Abstract Syntax Tree (AST) schema unifying syntactic representations across 10 programming languages using Tree-sitter grammars compiled into a shared dynamic library.
- Built a six-stage parsing and normalization pipeline achieving a 99.99994% conversion rate across 7M+ source files with multi-stage preprocessing.
- Conducted cross-language structural correlation analysis via cosine similarity matrices and PCA projection of schema embeddings, demonstrating consistent syntactic alignment.

MalCodeAI: AI-Powered Malicious Code Detection with Explainability Jan. 2025 – May 2025

- Developed a two-stage pipeline for language-agnostic malicious code detection using fine-tuned Qwen2.5-Coder-3B-Instruct with LoRA on MLX, assigning CVSS scores and providing exploit details and fixes.
- Conducted systematic hyperparameter ablations (iterations, trainable layers, learning rate) across both pipeline stages, reducing validation loss from 0.865 to 0.399 and from 1.639 to 0.199 respectively.
- Achieved a mean usefulness score of 8.06/10 from developer feedback across 15 evaluations.

TECHNICAL SKILLS

Programming: Python, C++, Java, SQL, Bash

Machine Learning & AI: PyTorch, TensorFlow, Hugging Face Transformers and TRL, Reinforcement Learning (RL), Reinforcement Learning from Human Feedback (RLHF), Preference Optimization (GRPO/DPO), Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Graph Neural Networks (GNNs), AI Agents

Systems & Infrastructure: Linux, Git, Docker, Apache Spark, Hadoop, AWS, Google Cloud

Research Interests: Reinforcement Learning, AI Alignment, AI for Software Engineering and Security, Decision Making Under Uncertainty, Large Language Models, Graph Representation Learning, Explainable AI (XAI), Agentic AI Systems, Knowledge Graph Reasoning

RELEVANT EXPERIENCE

Graduate Assistant

The George Washington University, *Washington, DC* Sep. 2025 – Dec. 2025

- Supported an upper undergrad- & graduate-level Big Data and Analytics course (CSCI 4907/6444).
- Guided and graded students on assignments and projects, focusing on tools like Spark, Hadoop, and S3 Bucket, and programming in languages like Python, Java, and Linux-Bash.
- Assisted with analytical methods, including machine learning, and provided support for cloud-based and distributed systems projects.

Teaching Assistant

Navrachana University, *Vadodara, India* Jan. 2024 – May 2024

- Mentored undergraduate students through code review, grading, and structured feedback on 750+ lab reports.
- Assisted in designing rubrics, evaluating technical writing, and improving student understanding of test-driven development.
- Supported curriculum delivery along with faculty, focusing on code quality, documentation, and debugging.

NVIDIA Jetson AI Project Coordinator

Navrachana University, *Vadodara, India* Mar. 2022 – Jun. 2022

- Conducted workshops on CV and NLP using the Jetson Nano platform, introducing students to embedded AI and real-time inference.
- Mentored 5+ student research projects selected by the NVIDIA Deep Learning Institute.
- Promoted student research initiatives and hands-on learning in edge AI systems.

CERTIFICATIONS & HONORS

Sustainability Award (Complex Networks 2025 Conference)	2025
NVIDIA-Certified Associate: Generative AI LLMs	2025
GW SEAS Dean's Award Scholarship	2024
Runner-Up, Tinkerthon 2.0 Hackathon	2024
SSIP Innovation Grant (Project EzyCart)	2023
NVIDIA Jetson AI Specialist	2022